The Restricted Invertibility Conjecture in arXiv:1010.6090 Is Already Answered

Literature-status note — August 17, 2026

Status. `literature_already_answered`. This is an exact status identification, not a new mathematical result.

Source problem. Nikolski and Vasyunin state the following conjecture in Section 3.2, PDF page 19 [1]:

> For every bounded operator T on a Hilbert space H and every orthogonal basis (e_j) satisfying $\inf_j \|Te_j\|/\|e_j\| > 0$, there is a finite partition $\bigcup_{s=1}^r I_s = \mathbb{N}$ such that every restriction $T|_{H_{I_s}}$ is left invertible.

Immediately before the display, the source records the known implication

$$\text{positive Kadison–Singer solution} \quad \Longrightarrow \quad \text{RIC},$$

citing Casazza–Christensen–Lindner–Vershynin.

Answer. Marcus, Spielman, and Srivastava prove two theorems known to imply a positive solution of the Kadison–Singer problem [2, abstract and Corollary 1.5]. Their introduction explicitly includes the Bourgain–Tzafriri conjecture among the equivalent formulations of Kadison–Singer. Combining their positive solution with the implication already recorded by the source gives a positive answer to the displayed RIC.

The supporting authors were aware of this relationship: arXiv:1306.3969 names the Bourgain–Tzafriri conjecture in its equivalence list rather than merely supplying an unrelated theorem later recognized to apply.

Other source signal. Remark 3 of arXiv:1010.6090 says that the first Blaschke product constructed there is not known to have a noninvertible element exactly at its threshold. Theorem 2.5 later in the same paper constructs a refined product with exactly that endpoint property. This signal is a same-paper ask-and-answer passage, not a separate open problem.

Scope limitation. The paper also names stronger variants for Schauder bases (B-RIC) and summation bases (SB-RIC), and refutes SB-RIC. This note classifies only the formally displayed RIC. It makes no claim about the current status of B-RIC.

References

[1] N. Nikolski and V. Vasyunin, *Invertibility threshold for H^∞ trace algebras, and effective matrix inversions*, arXiv:1010.6090 (2010), especially PDF p. 19.

[2] A. W. Marcus, D. A. Spielman, and N. Srivastava, *Interlacing Families II: Mixed Characteristic Polynomials and the Kadison–Singer Problem*, arXiv:1306.3969v4; Ann. of Math. 182 (2015), 327–350.